DC motor: PWM and PID speed study

ME3202 control lab concept | Gajendra, 2301ME18

Abstract - short summary

This independent Python study compares a fixed PWM command with PID speed control on a normalized first-order motor model. It makes the control loop and its output visible for interview preparation. The plotted speeds are simulated values, not measurements from the IIT Patna lab motor.

Purpose and background

The ME3202 course covered DC motor PWM and closed-loop PID with Arduino, a driver and Simulink. This repository model isolates the control principle so target tracking can be reproduced without hardware.

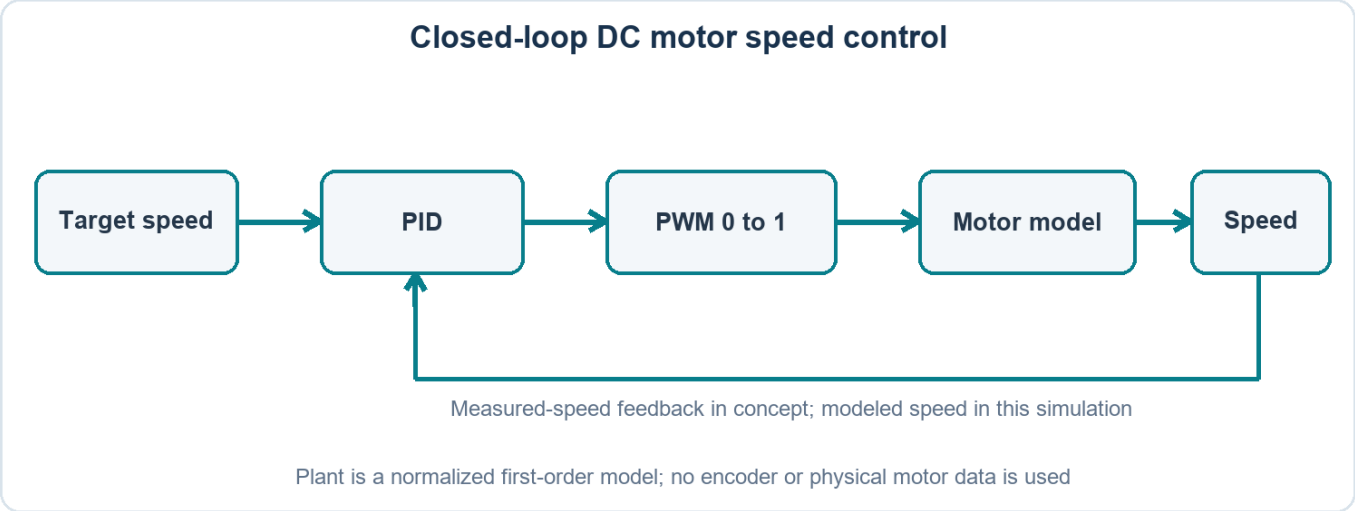


Figure 1. Simulated closed-loop structure. The diagram does not imply an encoder was used in this Python study.

Methods - how the study was carried out

- Use d(speed)/dt = (gain * PWM - speed) / time_constant with unitless speed and PWM.
- Apply fixed PWM = 0.45 to one copy and a PID command to another; clamp PID PWM to 0 through 1.
- Integrate with 0.02 s steps for 4.0 s using the documented default gains; compare against target 0.70.

DC motor: PWM and PID speed study | Work and results

What was done

The items below describe the work represented by this repository and the course context.

• Gajendra reports completing the corresponding motor control lab at IIT Patna.

• This repository adds an independent Python model, CSV output and tests for fixed-PWM versus PID behavior.

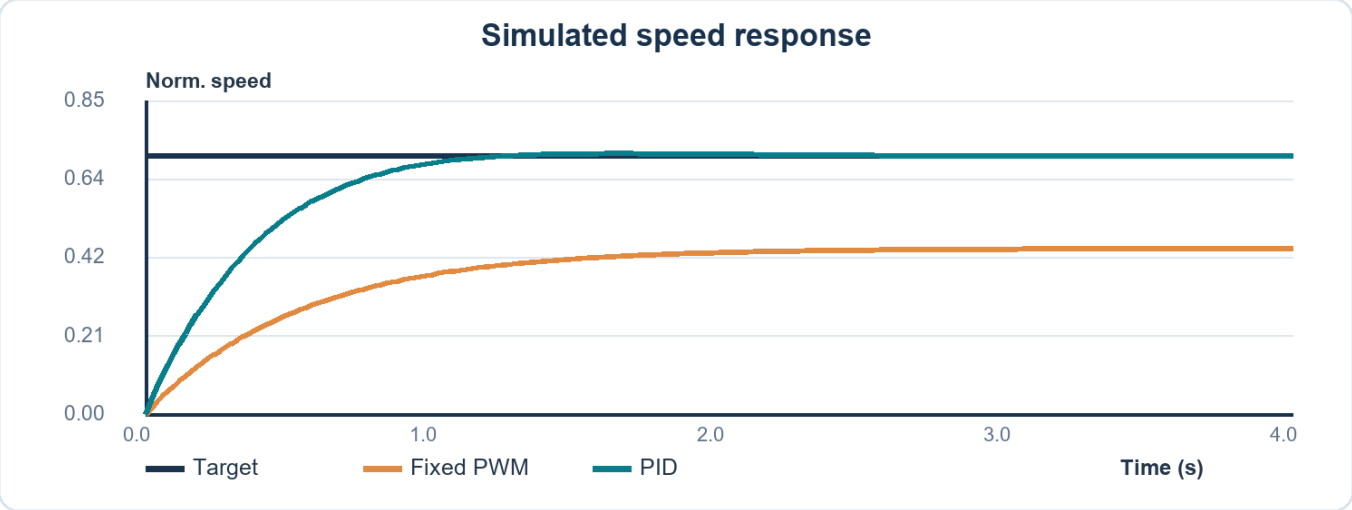


Figure 2. Speed trajectories generated by the repository code with default parameters.

Result and interpretation

At 4.0 s, fixed PWM gives normalized speed 0.450; PID gives 0.700 against target 0.700. With these parameters, integral action removes the persistent error present under the fixed command.

Limits and next validation step

The model has no measured motor constant, encoder noise, load disturbance, dead zone or direction reversal. For hardware validation, log commanded duty, measured RPM and load changes, then retune gains while checking saturation and safety.

Reproduce it

Run: python projects/dc_motor_pid/dc_motor_pid.py

Open project code on GitHub